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Graded Assignment: Module 4 Summative Assessment Submitted
Module 4 Summary

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

Module 4 Summative Assessment

Review Learning Objectives

Instructions

coach
Ready to review what you've learned before starting the assignment? I'm here to help.
Help me practice Let's chat

Assignment details
Due Nov 17, 11:59 PM IST
Attempts 1 every 8 hours, 2 remaining
Submitted Nov 16, 1:58 PM IST

Your grade 100%
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1. In the context of smoothing splines, what is 'overfitting'? **1 / 1 point**

- Fitting the spline too closely to the data, capturing noise rather than the underlying trend.
- A situation where the spline is not smooth enough.
- Fitting a spline to a dataset with insufficient data points.
- Using a spline with too few knots.

Correct
Correct! Overfitting occurs when the model captures noise instead of the true pattern. Try again

2. What is the main purpose of adding a regularization term to a machine learning model's loss function? **1 / 1 point**

- To reduce overfitting by penalizing more complex models.
- To enhance the model's speed and computational efficiency.
- To increase the model's prediction accuracy on the training data.
- To automatically select the most relevant features in the dataset.

Correct
Correct! Regularization aims to reduce overfitting by penalizing model complexity.

3. In the context of piecewise polynomial regression, what is a 'knot'? **1 / 1 point**

- A point where the polynomial pieces connect or join together.
- A point of maximum curvature in the polynomial model.
- The highest degree of the polynomial used.
- A parameter that controls the overall fit of the model.

Correct
Correct! Knots are the points that separate different polynomial segments in piecewise regression.

4. Why are smoothing splines preferred over polynomial regression in some cases? **1 / 1 point**

- They provide flexibility in fitting data and avoid problems like Runge's phenomenon in polynomial regression.
- They are always computationally less expensive.
- They are simpler to implement than polynomial regression.
- They can only be used with linear models.

Correct
Correct! Smoothing splines offer more flexibility and can avoid issues like oscillations at the boundaries.

5. In the context of machine learning models, how does the use of a kernel function typically impact model performance? **1 / 1 point**

- It simplifies the model to prevent overfitting.
- It directly reduces the training time of the model.
- It reduces the need for feature engineering.
- It allows models to capture more complex patterns by transforming data into a higher-dimensional space.

Correct
Correct! Kernel functions enable models to handle more complex data patterns by mapping data into a higher-dimensional space.

6. How does a cubic spline differ from a piecewise linear regression? **1 / 1 point**

- Cubic splines cannot fit non-linear data patterns.
- Cubic splines use higher-degree polynomials than piecewise linear regression.
- A cubic spline uses third-degree polynomials and ensures smoothness at the knots.
- A cubic spline is always a linear function.

Correct
Correct! Cubic splines use cubic polynomials and ensure continuity and smoothness at knots.

7. How does adding a penalty term based on the size of model coefficients affect a machine learning model? **1 / 1 point**

- It encourages the model to balance fitting the data with maintaining simpler model complexity.
- It leads to larger coefficients in the final model.
- It removes redundant features from the dataset.
- It increases the likelihood of the model fitting noise in the data.

Correct
Correct! Penalizing based on coefficient size helps balance data fit and model simplicity.